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Table of Contents

Introduction 4
Literature Review 4
Methodology 5
Data 5
Features 6
Data preparation 7
Exploratory analysis 9
Models 10
Experiments 11
Results and discussion 12
Run 1 12
Run 2 14
Run 3 15
Feature importance 16
Limitations and future work 16
Conclusion 17
References 18
Appendix A – Additional Tables and Graphs 19
Table 1: Variable definitions 19
Table 2: Run 1 – Detailed Results 20
Table 3: Run 2 – Detailed Results 20
Table 4: Run 3 – Detailed Results 21
Table 5: Combined comparison across all three runs 21
Table 6: Per-Certainty Breakdown - Run 2 vs Run 3 22
Table 7: Reweighted Comparison - Run 3 Adjusted to Run 2's Test Composition 23
Graph 8: Per Class F1 score by model and run 23
Appendix B – AI Usage Declaration 25

Abstract

This project explores Value Area shape classification for NQ futures using machine learning. Four classification models were trained on six statistical features of the volume distribution to distinguish between D, P, b and flat profiles. The best model – KNN – reached 97.9% accuracy on clearly defined shapes.

Introduction

This project focuses on volume profile shape classification using machine learning. I chose this topic because it is something I would like to solve for my own algorithmic trading projects. While there is a lot of literature on market profile theory and related trading concepts, I could not find any published research that specifically applies machine learning to classify volume profile shapes, which makes this research valuable to other people interested in this area as well.

Literature Review

Market profile is a concept in technical analysis introduced by J. Peter Steidlmayer (CBOT, 1985) and later formalised in his book Steidlmayer on Markets (Steidlmayer and Hawkins, 2003). He observed that the intraday distribution of trading volume typically follows a bell-shaped Gaussian curve. The Value Area is generally defined as the price range containing 70% of the day's traded volume around the Point of Control - the price level with the highest traded volume. The 70% figure was derived from the concept of one standard deviation in a normal distribution, which covers approximately 68.3% of the data.

Steidlmayer also observed that not all sessions produce a normal distribution of volume. Sessions that do are called D-shape profiles, which indicate that the market is in balance and buyers and sellers agreed on a fair price. When the distribution is skewed toward the top of the range, it creates a P-shape profile - buyers were in control during the session and pushed price to higher levels, where it consolidated. On the other side, the distribution can also be b-shaped, which signals that sellers were more aggressive and fair price was found at lower levels. The last shape is the Flat profile, which means that neither buyers nor sellers dominated and no clear normal distribution was formed either. Although Steidlmayer discussed primarily time-at-price (TPO) concept, volume profile (volume-at-price) is using the same principles.

Why is this concept important for traders and those who want to understand and predict market movements? Gebka (2025) used quantile regression to investigate the causality

between trading volume and stock returns. He found that the effect of volume on future returns is not uniform across the return distribution - at high return quantiles, volume causes positive future returns, while at low return quantiles, the effect reverses and volume causes negative future returns. These findings suggest that volume patterns carry predictive information about future price direction, which supports the relevance of my research on value area shapes.

Another piece of evidence for the importance of market profile shapes comes from Dalton et al. (2013). Their work focuses more on market profile patterns related to initial balance movements, but their six market day types still support the relevance of my VA shape research. They distinguish several recurring day structures, including the Normal Day, Normal Variation Day, Trend Day, Double-Distribution Trend Day, Non-trend Day, and Neutral Day. Although these categories are not identical to value-area or profile-shape classifications, they support the broader view that intraday profile structure contains useful information about market behaviour.

Methodology

Data

For this project I used my own dataset. I manually marked daily volume profiles for NQ (Nasdaq-100) futures across the full 2025 year and the first quarter of 2026 (till March 31), which gives me 312 labelled sessions in total. Each session is defined as one CME trading day. Since there is no exact recipe for marking volume profile shapes, my dataset is inherently subjective. However, I used Steidlmayer's and Dalton's principles described above. For each labelled session, I also have a precomputed volume profile with 1-point price bins, built directly from tick data. These profiles are the base for computing all features described below.

Besides the shape itself, I also marked each session with a certainty rating: A for textbook-clear cases, B for profiles that are recognisable but not perfect examples, and C for cases

where I was not confident in the shape assignment. The rating is used later to filter the dataset and to design the experimental runs.

Features

As I described earlier, market profile shape recognition is derived from the statistical concept of Gaussian distribution. Therefore, the first set of features uses standard statistical concepts applied to the volume-at-price distribution, with each feature designed to help the model distinguish between specific class pairs.

Altaf et al. (2022) demonstrated that statistical distribution features - mean, RMS, skewness and kurtosis – can be used for bearing fault diagnosis task using vibration signals. Although their work focuses on signal processing rather than distribution shape analysis, the same statistical features that distinguish fault types in vibration signals can also characterise differences in volume distribution shapes. Weighted skewness should help distinguish between P and b shapes. Excess kurtosis should mainly separate flat profiles from more peaked ones. Normalised POC position acts as a central tendency measure, also relevant for the P vs b distinction.

Shannon entropy has been demonstrated as an effective distribution feature across diverse classification tasks (Sepúlveda-Fontaine and Amigó, 2024). I computed the normalised Shannon entropy of the volume-at-price distribution. Similarly to kurtosis, entropy should help me distinguish between flat profiles (high entropy, volume spread evenly) and all other shapes.

Bimodality Coefficient is another feature derived from skewness and kurtosis. It can statistically distinguish between unimodal and multimodal distributions. Pfister et al. (2013) discussed BC and other tools for this purpose, and stated that BC has merit for assessing bimodality, although none of the tested statistics is “perfectly sensitive and specific at the same time”. Flat profiles tend to be more multimodal than other profile types, which is what this feature should capture. The formula I used was simplified (omits

the sample size correction part $(n-1)^2$ / ((n-2) x (n-3))), because at our bin counts (~300 - 400 per session), this correction is negligible.

Finally, I added Value Area width ratio, which measures what fraction of the session price range is covered by the 70% Value Area. A narrow VA ratio indicates concentrated profiles (D, P, b), while a wide ratio indicates flat profiles where volume is spread across most of the session range.

The final feature list with relevant formulas is shown below (variables are explained in Appendix A):

1. Weighted Skewness $= \frac{\sum_i w_i (x_i - \mu)^3}{W \cdot \sigma^3}$; where Weighted mean $\mu = \frac{\sum_i w_i x_i}{W}$ and Weighted standard deviation $\sigma = \sqrt{\frac{\sum_i w_i (x_i - \mu)^2}{W}}$
2. Excess Kurtosis $= \frac{\sum_i w_i (x_i - \mu)^4}{W \cdot \sigma^4} - 3$
3. Normalised POC position $= \frac{x_{POC} - x_{min}}{x_{max} - x_{min}}$
4. Value Area Width Ratio $= \frac{x_{VAH} - x_{VAL}}{x_{max} - x_{min}}$
5. Normalised Shannon Entropy $= \frac{-\sum_i p_i \log_2 p_i}{\log_2 N}$, $where\ p_i = \frac{w_i}{W}$
6. Bimodality Coefficient $= \frac{skewness^2 + 1}{kurt_{excess} + 3}$

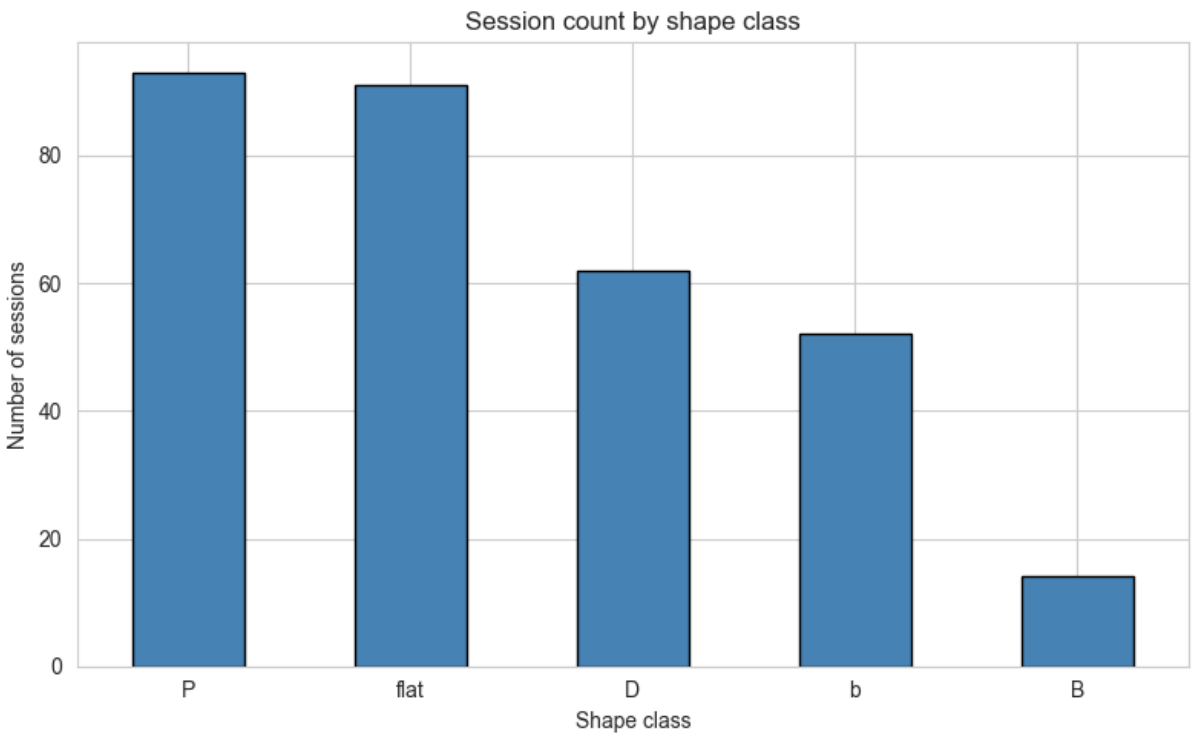
Data preparation

I decided to remove all C-certainty profiles from the dataset (14 in total). C-certainty profiles are the ones I was not confident in when labelling and including them would only inject noise into training without contributing meaningful signal.

Another concern is that the labelled dataset originally contains five shape classes. The fifth class is B-shape (double distribution), which has only 14 samples in the full dataset

and only 4 of those are A-certainty. I decided to drop all B-shape samples as well because the sample size is too low for reliable classification.

For further processing, I continue with 286 samples across 4 classes (D, P, b, flat) and two certainty ratings (A and B). The final class distribution is - flat 87, P 90, D 59, b 50. As we can see, classes are imbalanced – P-shape profiles cover 31% of the dataset, while b is the smallest at 17%. This imbalance is handled via SMOTE during training (see Models section below).



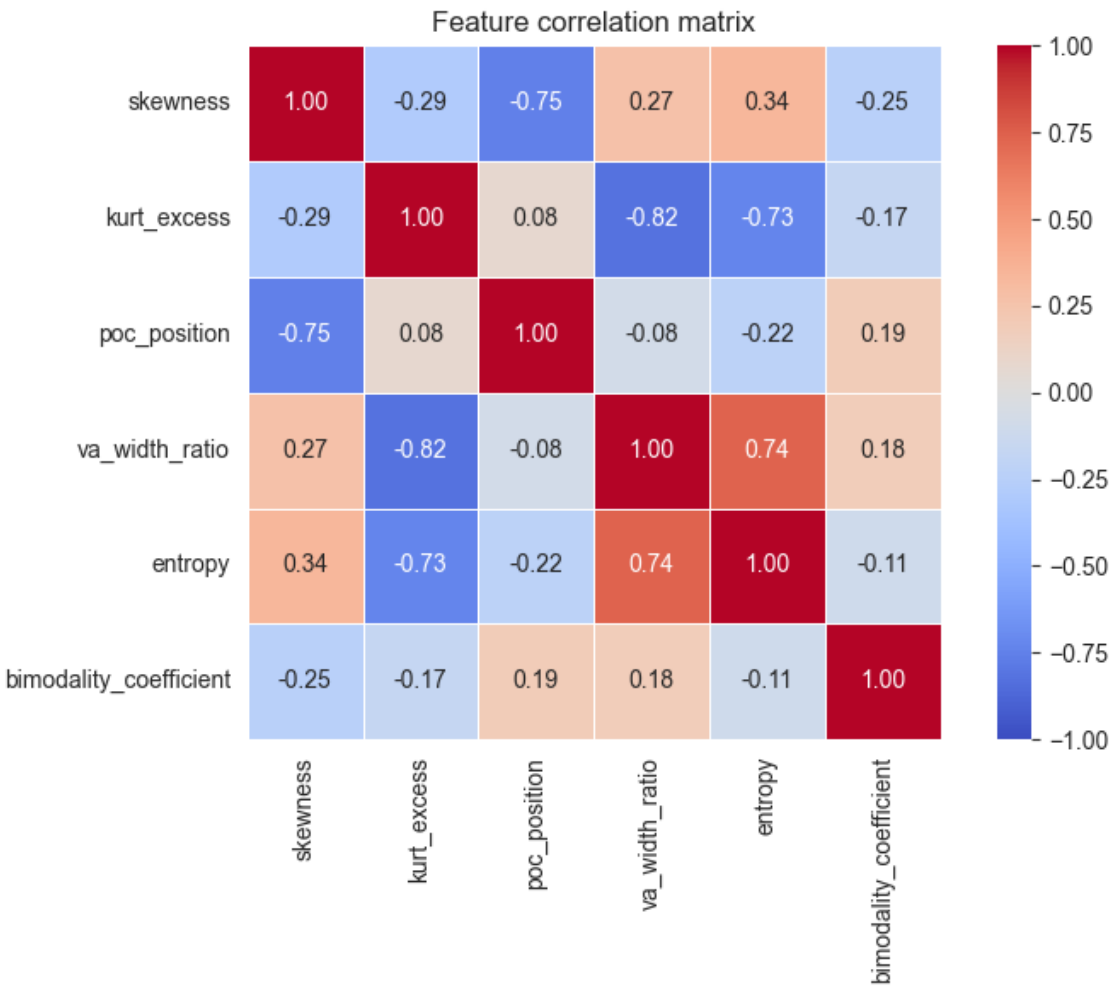
For the train/test split, I used an 80/20 stratified split, which I considered a good balance at this dataset. Stratification handles the class imbalance. The split is performed separately for each experimental run (see Experiments section).

Exploratory analysis

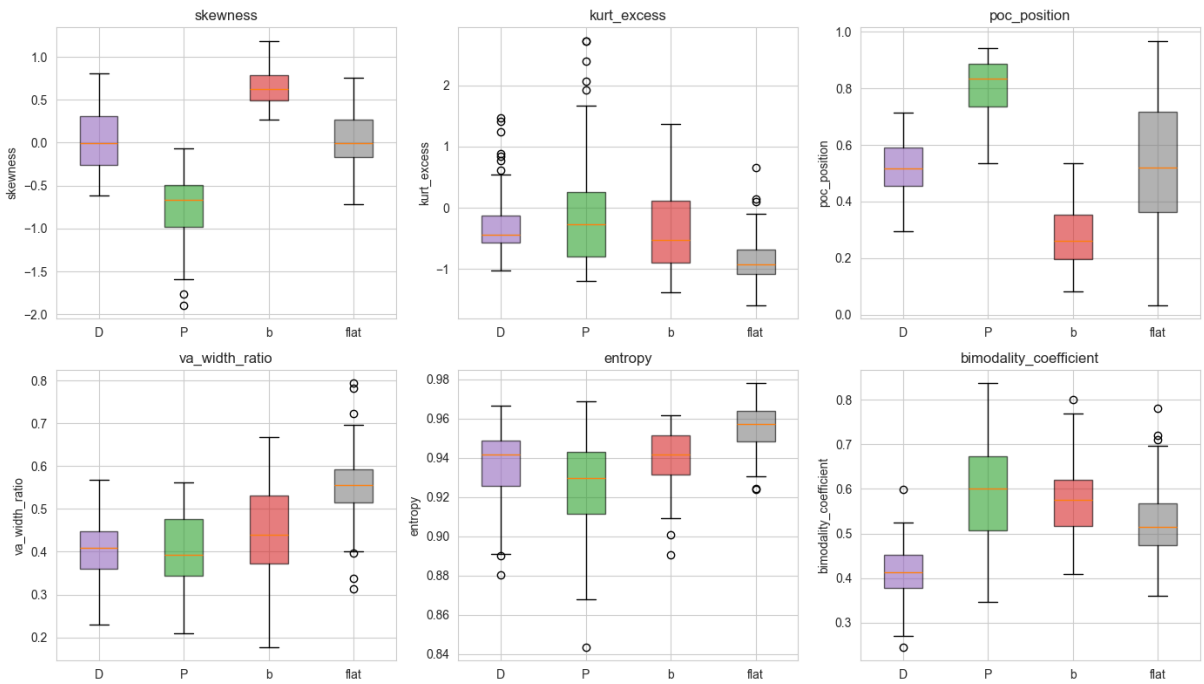
I also conducted basic exploratory analysis to check for any correlation concerns and to see some descriptive statistics for the features across different classes.

After computing the Pearson correlation matrix, I could flag 3 higher correlations. The highest is negative correlation of -0.82 between excess kurtosis and VA width, and both of these have also higher correlation (above |0.7|) with entropy. This is expected as they should all predict mainly the distinction between flat profiles and the rest. Another higher correlation has been observed between skewness and POC position (-0.75), which is also expected as they both should predict P and b shapes.

None of these correlations were assessed as too high, so no adjustments were made to avoid multicollinearity.



I also plotted box plots summaries, showing the median, interquartile range and outliers for each feature across the four classes. This shows visually how different features vary across classes and their informative power for distinguishing them. The box plots mostly confirm my original hypotheses - skewness and POC position show clear distinction between P, b and D, while VA width, entropy and kurtosis excess have differences between flat and D-shape, which is the desired behaviour.



Models

For my classification task, I chose 4 classification models for comparison - Random Forest, XGBoost, Support Vector Machine with RBF kernel, and K-Nearest Neighbours. By choosing those I will cover different algorithmic families - tree-based ensembles (RF and XGBoost), margin-based classifier (SVM), and instance-based learning (KNN).

For each model, I defined a small hyperparameter grid and ran grid search with stratified 5-fold cross-validation on the training data. The grids cover the 2-3 most impactful parameters per model - n_estimators and max_depth for Random Forest, n_estimators,

max_depth and learning_rate for XGBoost, C and gamma for SVM, n_neighbors and weights for KNN. The grid search was optimised against macro F1, which is a standard scoring metric for multiclass classification with imbalanced datasets because it treats all classes equally regardless of their sample size (Grandini et al., 2020).

Both SVM and KNN are distance-based and therefore sensitive to feature scale, so I applied StandardScaler to their inputs. Random Forest and XGBoost are scale-invariant and do not need it. To handle the class imbalance in my dataset, I applied SMOTE which generates synthetic samples for the minority classes by interpolating between existing ones (Chawla et al., 2002).

Experiments

I decided to run the models three times, to test different aspects of the problem based on the samples' certainty rating.

Run 1 uses A-certainty samples only for both training and test. The goal is to find out what the potential of the models is when I give them clear examples of the classes, which shows if the features themselves are good enough.

Run 2 uses both A and B rated samples for both training and test, with a fresh train/test split from the A+B pool. This shows how the results change when I include also ambiguous samples that naturally appear in the real world.

Run 3 uses the A-only training set from Run 1 but evaluates on a test set that combines the held-out A samples from Run 1 with all B samples. This shows how a cleanly-trained model generalises to a more realistic, mixed-quality test distribution.

Because Run 2 and Run 3 use test sets with different A/B compositions, their overall metrics are not directly comparable. To address this, I split the Run 2 and Run 3 predictions by certainty and computed metrics separately on each portion. I also reweighted Run 3's per-certainty accuracies to match Run 2's A/B ratio, producing an estimate of Run 3's score on Run 2's test composition. This is not an ideal apples-to-

apples comparison, as the actual test samples are still different, but it provides a reasonable estimate and decent comparison of these two runs.

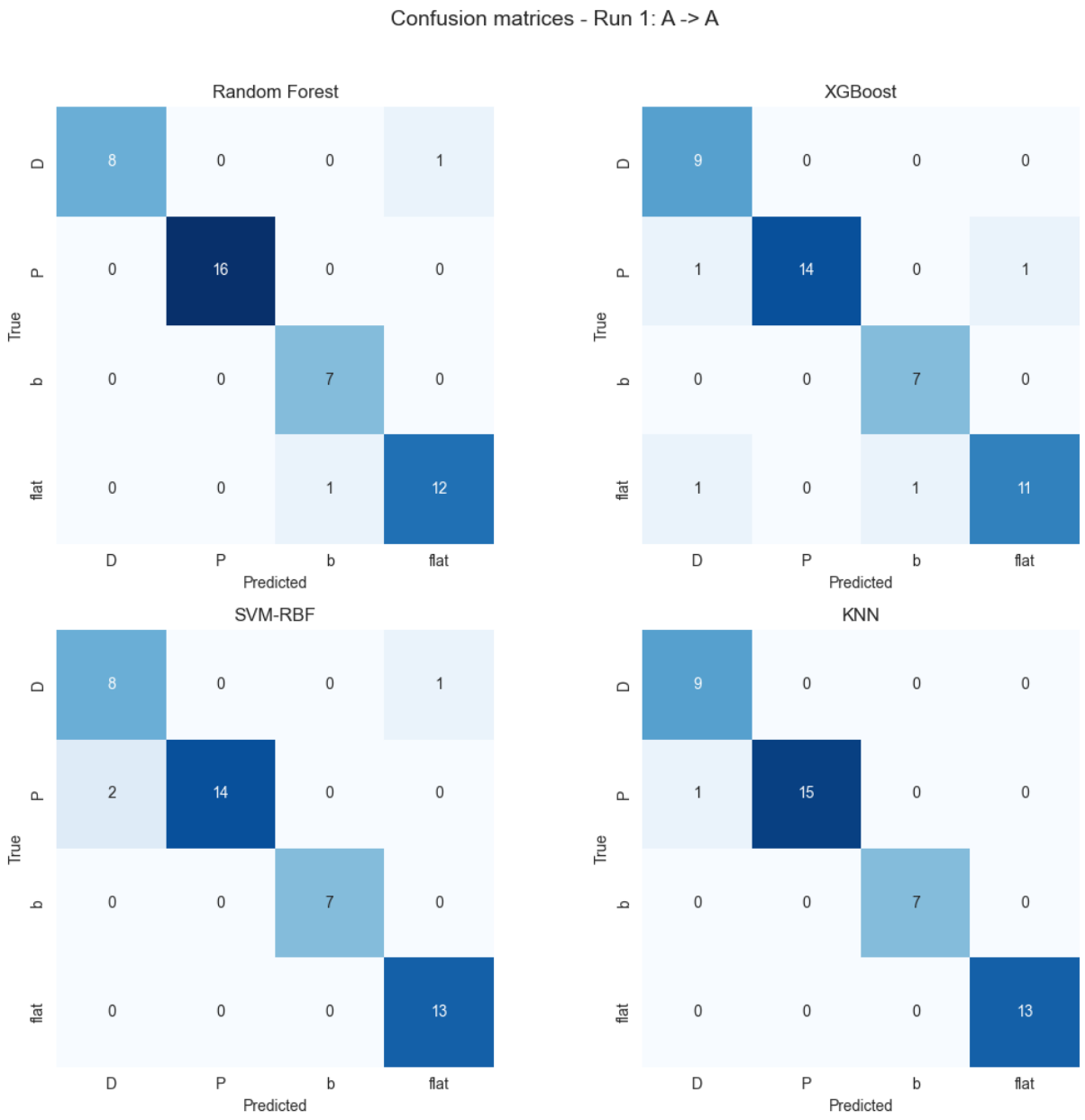
Results and discussion

Overall, Run 1 produced the best results, which was expected since the dataset consisted of "textbook" clear shapes. KNN appeared to be the best model, with 97.8% accuracy and 97.9% F1 Macro score. Run 2 included also the ambiguous B samples, so it is not surprising that it scored lower across all models. Run 3 was expected to be somewhere in between, which has been confirmed after I adjusted the results to be comparable. Detailed results for each run are in Appendix A.

Run 1

In Run 1, KNN (n_neighbors = 3, weights = uniform) had the best performance, around 2.2 percentage points above the second-best Random Forest. Note that since my test set has only 45 samples, the differences between models are within a range of 1-3 errors, which I do not consider a reliable difference to say KNN is definitely the best.

When I look at the confusion matrices, the important thing is that none of the models made any confusion between P and b shapes, which are on opposite sides of the spectrum. Small confusion between flat and other shapes was expected, since the boundary between them is often subtle.

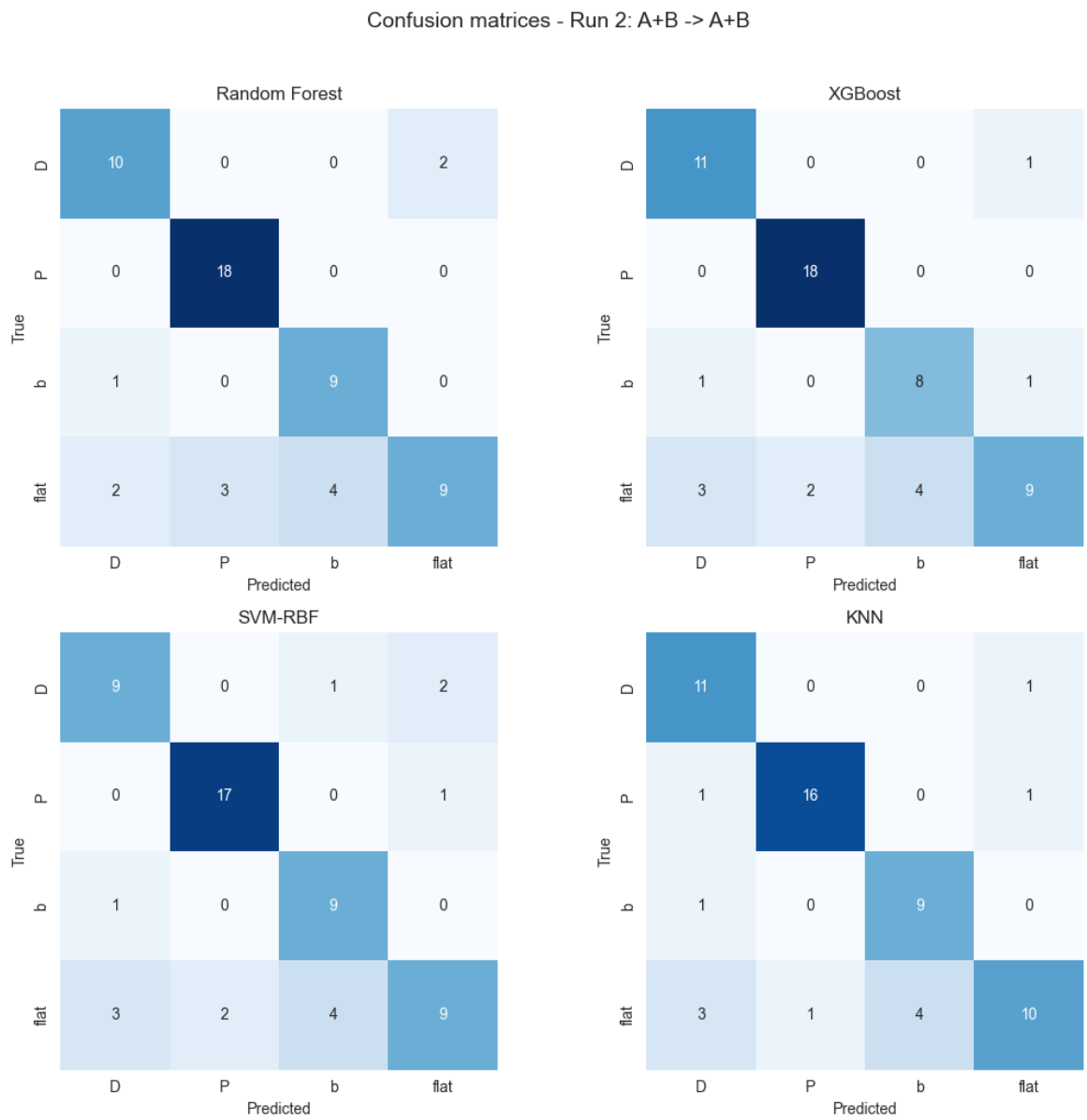


From the results, we can say that when we give any of the models reliable data, they are all able to get very good results, which I interpret as confirmation that the features chosen are relevant.

Run 2

In Run 2, KNN, Random Forest and XGBoost got exactly the same accuracy (79.3%), with only a tiny difference in Macro F1 score (all ~78%). The only model that stayed behind was SVM, with ~75%. This can be explained by the grid search selecting the highest C (10) as the optimal parameter, which forces SVM to fit the training data more precisely and increases the risk of overfitting.

We can see that the results are significantly worse than in Run 1, which was expected because the data included B-certainty samples. However, 79.3% accuracy still suggests relatively reliable potential for real use on real-time imperfect data.



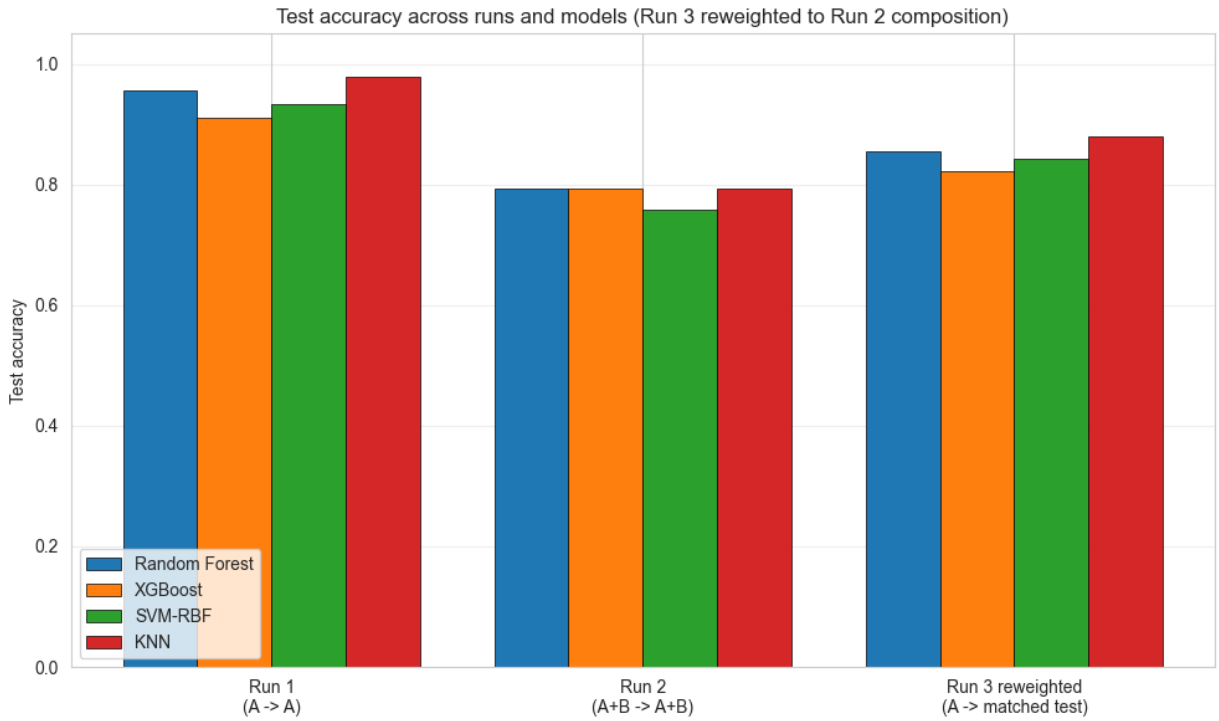
From the confusion matrices, we can see that, similarly to Run 1, the most problematic shape was flat, which is not as alarming as it would be if the errors were distributed across other classes.

Run 3

As mentioned in the Methodology section, Run 3 was conducted with an A-only training set and an A+B test set. The test set was composed of 45 A samples and 61 B samples, which by itself suggests worse results. The best model, KNN, scored 74.5%, which is still relatively strong considering the composition of the test set.

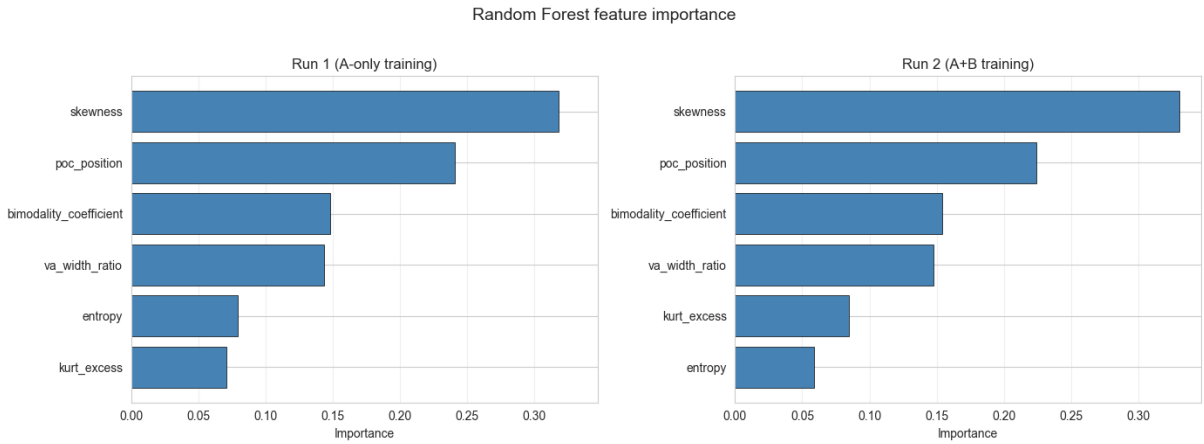
After reweighting Run 3's per-certainty accuracy to match Run 2's test composition, I was able to compare Run 2 and Run 3 results more reliably. Although it is not a direct apples-to-apples comparison, I think it is a decent estimate for this purpose. The reweighted accuracy jumped to 88% for KNN, with all other models landing around 82-85.5%.

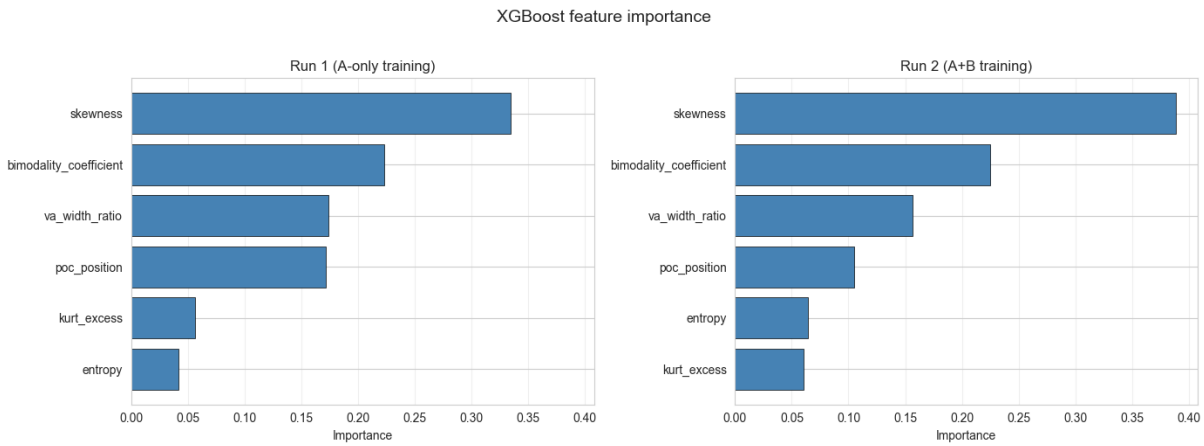
Based on this, I can say that training on A-only textbook samples makes more sense and delivers more reliable results - the reweighted accuracy beat Run 2 across all four models, with the best KNN making 9% percentage points difference.



Feature importance

Looking at Random Forest feature importance across Run 1 and Run 2, the most important features were Skewness, Normalised POC position, Bimodality Coefficient and VA width ratio, while excess kurtosis and entropy contributed the least. Skewness and POC position confirmed their expected role as the main P vs b discriminators. We can also see that BC and VA width ratio captured more information for flat vs concentrated distinction than excess kurtosis and entropy, which is an interesting insight.





Limitations and future work

There are a few limitations of my research that I would like to flag. First, the dataset is relatively small - 286 samples across 4 classes, which limits the statistical power of the results. A larger labelled dataset would give more reliable estimates.

Second, the dataset is subjective, as I marked all the profiles myself. Having multiple annotators would strengthen the validity of the labels, but this was out of scope for this project.

Third, I only tested the models on NQ futures. The features are designed to be market-agnostic, therefore they theoretically should work on other markets similarly, but proper testing is needed to prove this proposition.

Finally, B-shape (double distribution) was dropped due to small sample size. With more labelled data, this class could be added back to make the classification more complete.

Conclusion

I conducted a volume profile shape detection research - something that does not have any online available precedent. I demonstrated that from six statistical features of the volume distribution, the four chosen classification models can reliably distinguish between D, P, b and flat profiles, reaching up to 97.8% accuracy on clearly defined shapes. The models performed very similarly, with KNN scoring the best across all three runs, which makes it slightly more recommended than the other three. Training on high-certainty samples was found to deliver stronger results than training on the full dataset including ambiguous samples, which is a useful finding for anyone building automated shape recognition into algorithmic trading systems.

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Appendix A – Additional Tables and Graphs

Table 1: Variable definitions

Symbol	Meaning
x_i	Price level at bin i
w_i	Volume at price x_i
W	Total session volume, $W = \Sigma_i w_i$
N	Number of non-zero price bins
μ	Volume-weighted mean price
σ	Volume-weighted standard deviation of prices
x_{POC}	Point of Control - the price with the highest volume
x_{VAL}, x_{VAH}	Value Area Low and High (70% volume bounds)
x_{min}, x_{max}	Minimum and maximum price in the session
p_i	Normalised probability at bin i, $p_i = w_i / W$

Table 2: Run 1 – Detailed Results

Train size: 180 | Test size: 45

Model	CV macro F1	Test accuracy	Test macro F1	Best hyperparameters
Random Forest	0.8338	0.9556	0.9494	max_depth = None, n_estimators = 100
XGBoost	0.8382	0.9111	0.9117	learning_rate = 0.02, max_depth = 5, n_estimators = 200
SVM-RBF	0.8516	0.9333	0.9346	C = 1, gamma = 0.1
KNN	0.8251	0.9778	0.9788	n_neighbors = 3, weights = uniform

Table 3: Run 2 – Detailed Results

Train size: 228 | Test size: 58

Model	CV macro F1	Test accuracy	Test macro F1	Best hyperparameters
Random Forest	0.8266	0.7931	0.7816	max_depth = 5, n_estimators = 100
XGBoost	0.8255	0.7931	0.7775	learning_rate = 0.02, max_depth = 5, n_estimators = 300
SVM-RBF	0.8412	0.7586	0.7472	C = 10, gamma = scale
KNN	0.8305	0.7931	0.7873	n_neighbors = 5, weights = distance

Table 4: Run 3 – Detailed Results

Train size: 180 (same as Run 1) | Test size: 106 (45 A + 61 B)

Run 3 reuses the fitted models from Run 1, so CV macro F1 and best hyperparameters are identical to Table 2. Only the test metrics change because the test set is different.

Model	Test accuracy	Test macro F1
Random Forest	0.7170	0.7231
XGBoost	0.6981	0.7076
SVM-RBF	0.7170	0.7237
KNN	0.7453	0.7517

Table 5: Combined comparison across all three runs

Sorted by test macro F1 within each experiment

Experiment	Model	CV macro F1	Test accuracy	Test macro F1	Test weighted F1
Run 1: A → A	KNN	0.8251	0.9778	0.9788	0.9780
Run 1: A → A	Random Forest	0.8338	0.9556	0.9494	0.9556
Run 1: A → A	SVM-RBF	0.8516	0.9333	0.9346	0.9340
Run 1: A → A	XGBoost	0.8382	0.9111	0.9117	0.9113
Run 2: A+B → A+B	KNN	0.8305	0.7931	0.7873	0.7881

Experiment	Model	CV macro F1	Test accuracy	Test macro F1	Test weighted F1
Run 2: A+B → A+B	Random Forest	0.8266	0.7931	0.7816	0.7795
Run 2: A+B → A+B	XGBoost	0.8255	0.7931	0.7775	0.7806
Run 2: A+B → A+B	SVM-RBF	0.8412	0.7586	0.7472	0.7497
Run 3: A → A+B	KNN	0.8251	0.7453	0.7517	0.7402
Run 3: A → A+B	SVM-RBF	0.8516	0.7170	0.7237	0.7152
Run 3: A → A+B	Random Forest	0.8338	0.7170	0.7231	0.7167
Run 3: A → A+B	XGBoost	0.8382	0.6981	0.7076	0.6915

Table 6: Per-Certainty Breakdown - Run 2 vs Run 3

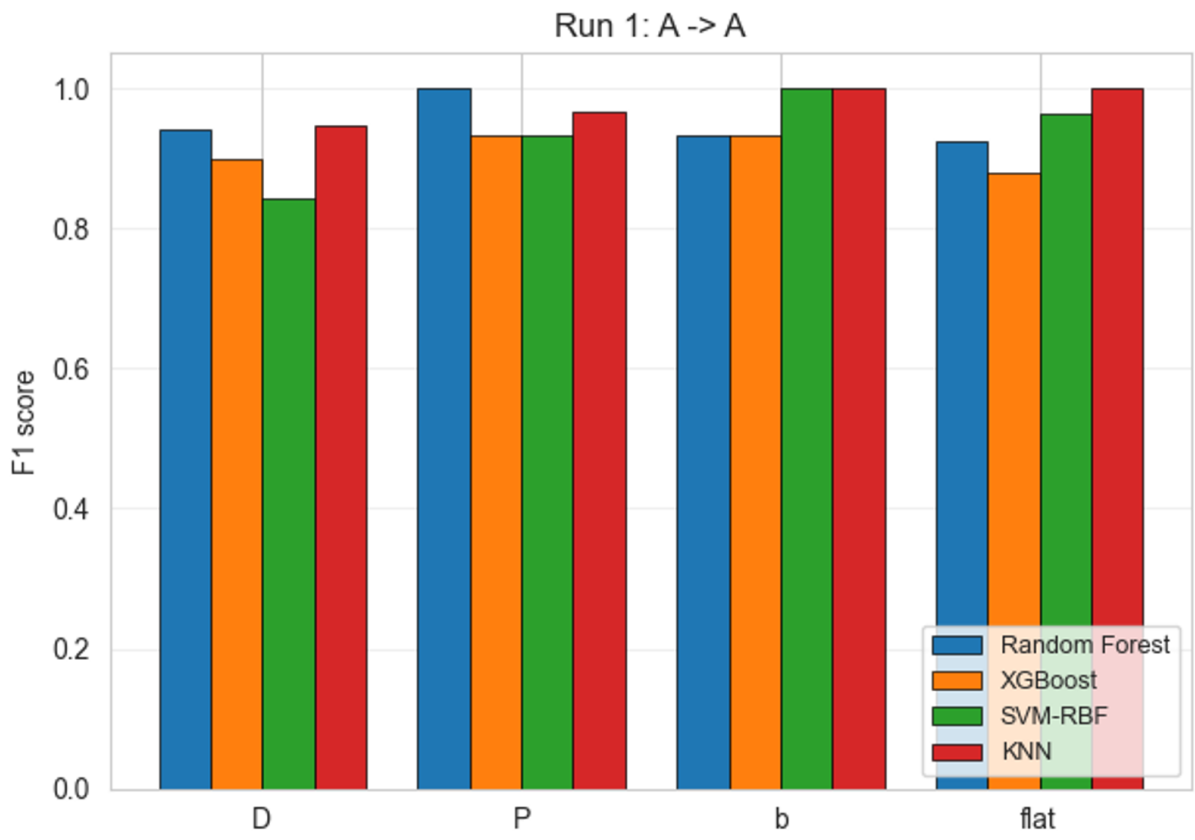
Model	Run 2 A acc	Run 2 B acc	Run 3 A acc	Run 3 B acc	A gap (R2-R3)	B gap (R2-R3)
Random Forest	0.8636	0.5714	0.9556	0.5410	-0.0919	+0.0304
XGBoost	0.8409	0.6429	0.9111	0.5410	-0.0702	+0.1019
SVM-RBF	0.8409	0.5000	0.9333	0.5574	-0.0924	-0.0574
KNN	0.8409	0.6429	0.9778	0.5738	-0.1369	+0.0691

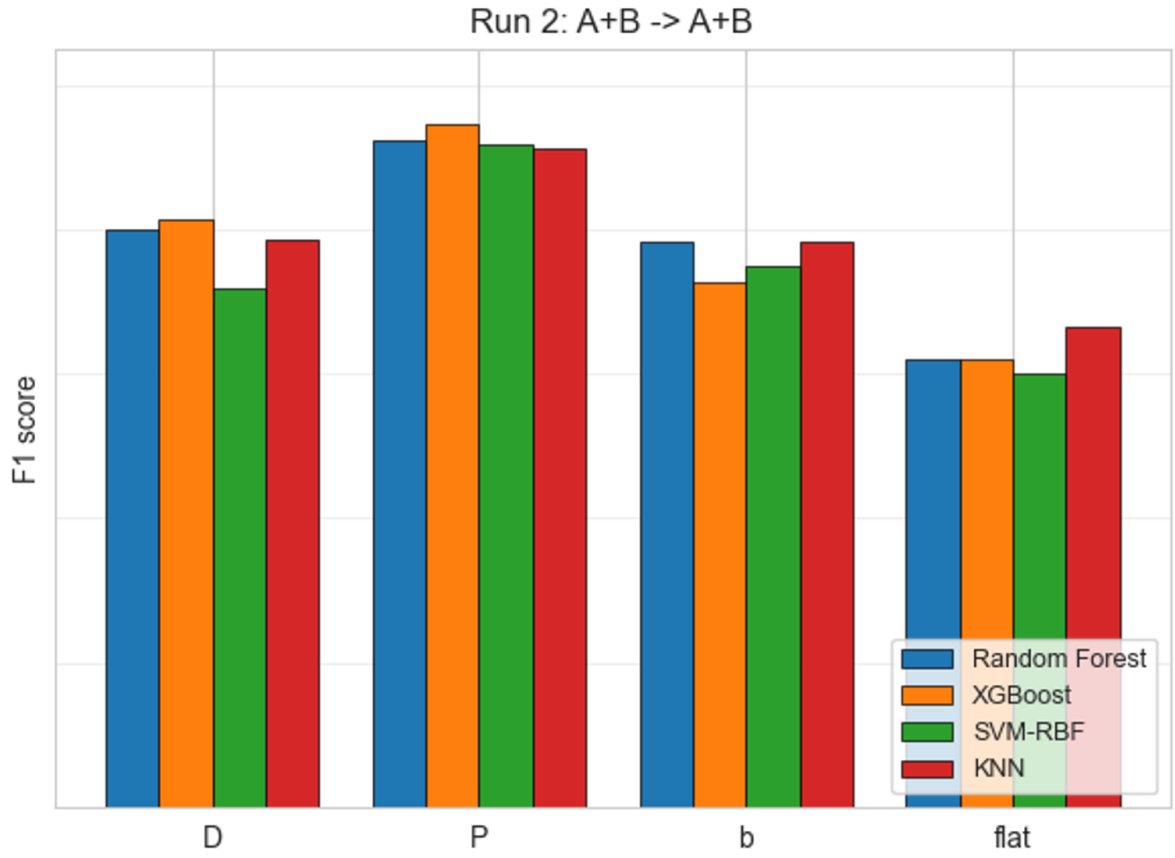
Table 7: Reweighted Comparison - Run 3 Adjusted to Run 2's Test Composition

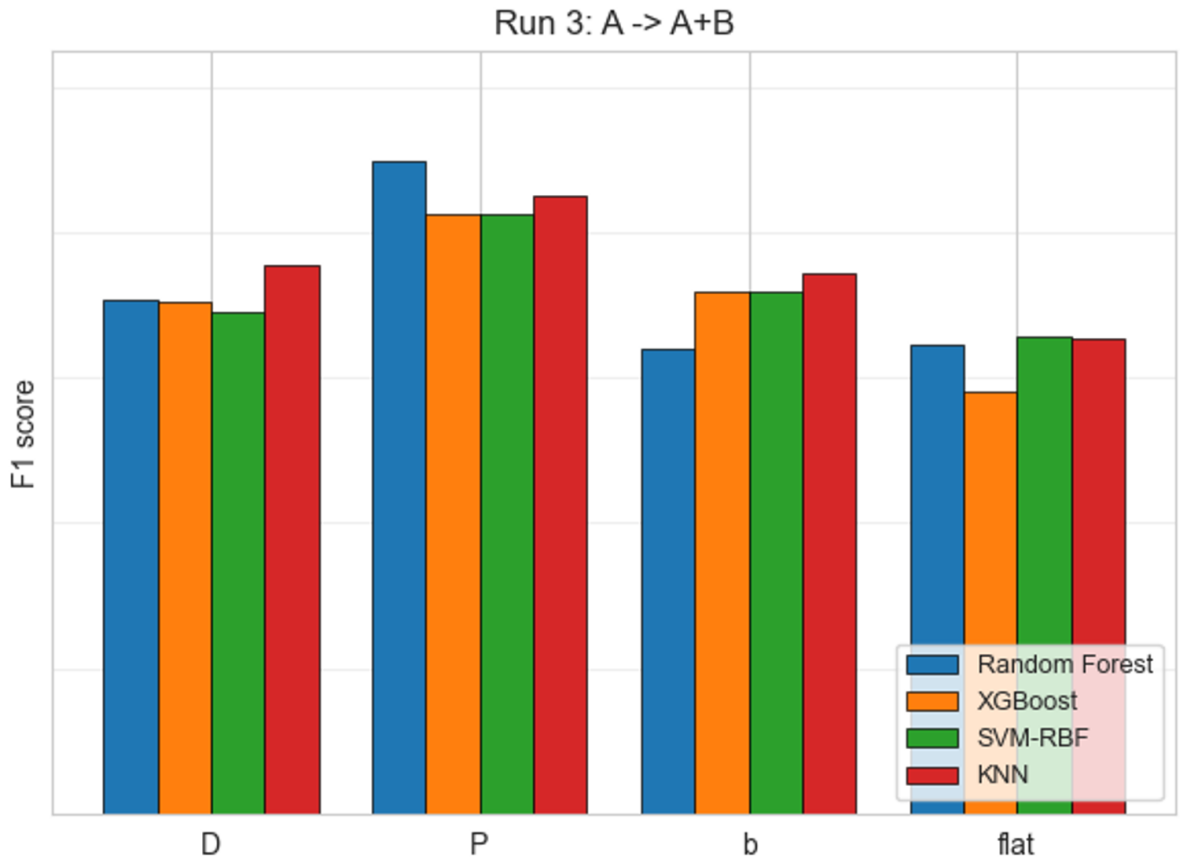
Model	Run 2 accuracy	Run 3 accuracy (reweighted)	Acc gap (R3-R2)
Random Forest	0.7931	0.8555	+0.0624
XGBoost	0.7931	0.8218	+0.0287
SVM-RBF	0.7586	0.8426	+0.0840
KNN	0.7931	0.8803	+0.0872

Graph 8: Per Class F1 score by model and run

Run 3 original, not reweighted




Run 2: A+B -> A+B
F1 score
D
P
b
flat
Random Forest
XGBoost
SVM-RBF
KNN


Run 3: A -> A+B
F1 score
D
P
b
flat
Random Forest
XGBoost
SVM-RBF
KNN

Appendix B – AI Usage Declaration

I declare that I used Claude Opus 4.6 to assist with the completion of this assignment. I used it for the following purposes:

1. Assistance with coding – I used Claude to draft some parts of the code, including comments, based on my instructions. All Gen AI-generated code that I used was then checked and corrected by me to ensure that the code and computations were correct and functional. It also gave me valuable experience in using AI for coding purposes, which is considered the inevitable future of programming.
2. Grammar and spelling checks, rewording of unnatural phrasing, and other language-related aspects of the report.
3. Report summaries – Claude was used to summarise research papers, which significantly sped up the assessment of whether they were relevant to my topic or not. When I chose a paper to be included in my work, I read the relevant sections myself, as I found that Gen AI often tends to hallucinate when working with complex research papers.

All AI suggestions that influenced this assignment have been reviewed, evaluated and verified by me before being incorporated.